How to Read CLIP Meets DINO / NoLA

Computer Vision Module Research Project
Based on CLIP Meets DINO / NoLA (BMVC 2025)

Proposed contribution: DINO-Neighbourhood Confidence for reliable label-free visual adaptation

Prepared for project development and supervisor review

How this guide is tailored to you

You already have practical experience with vision-guided robotics, ROS/ROS2, embedded systems, and deploying inspection systems. You therefore do **not** need a slow introduction to image classification or neural networks. The useful learning gap is more specific:

- how CLIP turns language into a classifier;
- why DINO features can be visually stronger than CLIP features;
- how pseudo-labels make label-free adaptation possible—and dangerous;
- how visual prompt tuning changes a frozen transformer;
- how to read the experimental evidence as a researcher rather than only reproducing code.

The recommended strategy is a fast theory pass, followed by close reading of the original PDF and a small set of marked exercises. Robotics is used for intuition, but the paper should be understood and presented as a **computer-vision paper**.

1. What you should understand by the end

After using this guide, you should be able to:

– explain NoLA's three stages without looking at the paper;

– write down the input and output of every major component;

– explain Equations 1–3 in plain language and with tensor dimensions;

– distinguish zero-shot, label-free, self-supervised, semi-supervised, and transductive learning;

– interpret Tables 1 and 3–7 without repeating the authors' claims uncritically;

– locate the paper's main ideas in the official code;

– identify the confidence-only selection weakness that motivates our DINO-Neighbourhood Confidence extension;

– design a fair experiment in which labels are used only for evaluation.

2. The paper in one paragraph

CLIP knows how class names and descriptions relate to images, but its image features can miss fine visual distinctions. DINO learns strong visual structure from images without labels, but it does not naturally know the target class names. NoLA first uses LLM-generated class descriptions and CLIP to assign confident pseudo-labels to some target images. Those pseudo-labels teach a small adapter on frozen DINO features, creating a DINO-based labelling network. That network then pseudo-labels weakly augmented images while visual prompt tokens are trained inside a frozen CLIP image encoder using strongly augmented versions. The goal is better target-dataset classification without any human image labels.

3. The central intuition using a robotics analogy

Imagine deploying a robot's camera in a new warehouse. You know the names of the categories it should recognise, but no warehouse images have been labelled.

- **CLIP is the semantic engineer:** it understands the class names and textual descriptions but may overlook subtle visual details in the new environment.
- **DINO is the visual inspector:** it groups visually similar objects well but does not know what the groups are called.
- **The CDE classifier is a richer specification sheet:** instead of only saying “forklift,” it uses many visual descriptions of a forklift.
- **Pseudo-labels are automatically generated inspection tags:** cheap, useful, and sometimes wrong.

- **The DINO adapter teaches the visual inspector the class vocabulary.**
- **Visual prompt tuning adjusts CLIP to the new camera domain without retraining the whole model.**

This analogy is only motivation. The actual research problem is label-free image-classification adaptation.

4. Vocabulary you must keep separate

Term	Meaning in this paper	Common confusion
Zero-shot classification	Classify using pretrained CLIP and class text, without target-image training	It does not mean the class names are unknown
Label-free adaptation	Train on target images without using their human labels	The class vocabulary and LLM descriptions are still available
Self-supervised learning	Learn visual representations from images without semantic labels, as DINO does	DINO pretraining and NoLA target adaptation are different stages
Pseudo-label	A model-generated class target used as if it were a label	It is not ground truth and may be confidently wrong
Transductive setting	The method adapts using the unlabeled target-image collection	This differs from pure zero-shot evaluation on unseen target images
VLM	Vision-language model, here mainly CLIP	DINO is a visual SSL model, not a VLM
Prompt tuning	Optimise a small set of input tokens while freezing most model weights	It is not the same as writing natural-language prompts by hand
CDE	Class Description Embedding classifier built from many descriptions per class	It is more than the single template “a photo of a {class}”
DL network	DINO-based labelling network: frozen DINO plus a trained adapter	“DL” here does not simply mean deep learning
TAAL	Name used in the code for the DINO encoder and its adapter	Treat it as the implementation of the paper's labelling network

5. Minimum theory before reading the method

5.1 CLIP as a classifier

CLIP has an image encoder and a text encoder. Both output vectors in a shared embedding space. For each class, a text such as “a photo of a satellite image of a forest” is encoded into a class vector. An image is encoded into an image vector. Cosine similarity between the image vector and each class vector becomes the classification score.

The important weakness is that **semantic alignment is not identical to fine-grained visual discrimination**. CLIP may know what “Siamese cat” means while still producing less separable visual features than an image-only self-supervised model.

5.2 DINO

DINO trains a student and teacher vision transformer using self-distillation across different image views. It does not need class labels. Its features often preserve object structure and fine visual similarity well.

For this paper, you do not need to derive the full DINO loss. Remember only:

- DINO supplies a frozen image feature vector;

- those features are visually informative;
- a small adapter is needed to map them to the target classes.

5.3 Pseudo-label learning

If a model predicts class c with high confidence, that prediction can supervise another model. This scales without annotation, but errors can reinforce themselves. Confidence is only a model belief—not proof of correctness.

The key research issue is therefore not merely “can we generate pseudo-labels?” It is:

> Which pseudo-labels are trustworthy enough to influence adaptation?

5.4 Visual prompt tuning

A vision transformer converts an image into patch tokens. NoLA introduces learnable prompt tokens alongside those patch tokens. The transformer is mostly frozen; gradient descent updates the prompts and selected lightweight components. This is parameter-efficient and normally cheaper than full fine-tuning.

5.5 Weak and strong views

NoLA follows a FixMatch-like idea:

- a weakly transformed image is used to obtain a pseudo-label;
- a strongly transformed version is trained to predict the same class.

This encourages invariance to appearance changes while trying to preserve a reliable teacher signal.

6. The complete NoLA pipeline

Inputs

- an unlabeled collection of target-domain images;
- the finite target class names;
- multiple LLM-generated visual descriptions for each class;
- pretrained CLIP and DINO models.

Stage A — class-description embedding classifier

– Generate `K = 50` descriptions for every class.

– Encode every description with CLIP's frozen text encoder.

– Average the description embeddings within each class.

– Stack the class vectors into classifier matrix `phi`.

– Run target images through CLIP and compare image features with `phi`.

Output: class probabilities and pseudo-label confidence for each image.

Stage B — DINO-based labelling network

– Group images by their predicted class.

– Select the top-`k` confident examples per predicted class.

– Extract frozen DINO ViT-B/16 features for those images.

– Train a small adapter to predict the selected pseudo-labels.

The paper uses a hybrid sample budget:

- `k = 16` for sufficiently small datasets;

- otherwise about 20% of the average images per class;
- maximum `k = 512`.

Output: a DINO-based target-class predictor.

Stage C — DINO-assisted visual prompt learning

– Create weak and strong views of every target image.

– Use the DINO labelling network on the weak view to obtain a hard pseudo-label.

– Add learnable visual prompt tokens to CLIP's image encoder.

– Predict the class of the strong view.

– Update the visual prompts and CDE classifier with smoothed cross-entropy.

Output: a target-adapted CLIP classifier.

7. Equation walkthrough

Equation 1 — constructing the CDE classifier

For class `C`, encode its `K` descriptions and average them:

`phi_C = (1/K) sum_i F_t(omega_i^C; theta_t)`

Then concatenate all class vectors:

`phi = Concat[phi_1, phi_2, ..., phi_C]`

Use this dimension check:

- number of classes: `C`;
- descriptions per class: `K = 50`;
- CLIP embedding size: `d`;
- one encoded description: `[d]`;
- one averaged class vector: `[d]`;
- complete classifier: `[C, d]`.

The equation is simply **prototype construction in CLIP's text-image space**.

Equation 2 — prompted visual representation

`f_v^p = F_v(X^0; theta_v, theta_P)`

- `X^0` is the patchified image input;
- `theta_v` is the mostly frozen CLIP vision encoder;
- `theta_P` is the learnable visual prompt;
- `f_v^p` is the adapted image representation.

The scientific claim is that a small prompt can shift CLIP toward the target visual domain without full model fine-tuning.

Equation 3 — DINO-assisted prompt objective

The DINO-based labelling network predicts the class from the weak image view. The prompted CLIP model predicts from the strong view. Smoothed cross-entropy trains prompted CLIP to match the DINO-generated pseudo-label.

Read the equation as a teacher–student relation:

- **teacher target:** `h(g_s(X^0))` from frozen DINO plus adapter;

- **student prediction:** `phi^T F_v(X^s; theta_v, theta_P)`;
- **optimised variables:** visual prompts and the CDE classifier;
- **frozen visual backbones:** DINO and most of CLIP.

Do not get trapped by notation. Draw two image views, two branches, and one loss arrow.

8. How to read each figure and table

Figure 1

Treat it as motivation, not proof. It visually compares top-1 accuracy against other label-free methods. Verify the exact numbers in Table 1.

Figure 2 — the most important figure

Spend most of your first reading here. Cover the caption and reconstruct:

– where language enters;

– where unlabeled images enter;

– which model creates each pseudo-label;

– which parameters are trained in each stage;

– where DINO features influence CLIP adaptation.

If you cannot redraw Figure 2, reread Sections 3.2.1–3.2.3 before continuing.

Figure 3

The t-SNE plot suggests better class separation after adaptation. It is qualitative evidence only. t-SNE can distort global geometry, so it should support—not replace—accuracy and calibration measurements.

Table 1 — main result

Important reference values using CLIP ViT-B/32:

Dataset	CLIP	LaFTer	NoLA	Observation
EuroSAT	40.6	73.9	73.5	NoLA is slightly below LaFTer
DTD	42.9	46.1	56.1	Large NoLA improvement
OxfordPets	85.0	82.7	89.3	NoLA beats both
11-dataset average	64.7	73.0	76.6	Main aggregate claim

Read beyond the average. NoLA is not best on every dataset. Its value appears strongest where complementary visual structure helps, but that interpretation must be tested rather than assumed.

Table 2

Shows NoLA with CLIP ViT-B/16 and compares it with methods that do not all use exactly the same supervision. Note that AdaptCLIPZS uses 16 shots, so this is not a pure label-free comparison.

Table 3

This is central evidence for DINO's contribution:

- full DINO-based labelling: 80.5%;
- replacing DINO with CLIP: 77.8%;

• replacing the labelling network with CDE: 76.2%.

It supports complementarity, although six-dataset averages hide dataset-specific behaviour.

Tables 4 and 5

These show the contribution of trainable components and the incremental pipeline. Table 5 is especially useful for a presentation because it tells a simple story: CDE helps, DINO alignment helps, and prompt learning contributes the largest final gain.

Table 6

MAE and BEiT also improve over the displayed CLIP baseline, but DINO performs best. The broader conclusion is that the framework can benefit from image-only SSL representations—not that DINO is the only possible backbone.

Table 7

GPT-4o descriptions improve the ten-dataset average by 0.72 points over GPT-3.5 descriptions, but individual datasets sometimes decline. Better language descriptions are not uniformly better visual classifiers.

9. Claims, evidence, and what remains uncertain

Claim	Evidence supplied	What is still missing
NoLA improves average label-free accuracy	Table 1 across 11 datasets	Seed variance and confidence intervals
DINO contributes useful visual information	Tables 3 and 6	More analysis of when DINO helps or hurts
Prompt learning is important	Tables 4 and 5	Compute-normalised comparisons
Richer descriptions can help	Table 7	Sensitivity to LLM, prompts, and description filtering
The method is efficient	Lightweight trainable modules	Runtime, peak VRAM, energy, and full parameter accounting
Pseudo-labels are useful	Final accuracy and adapter ablations	Calibration, selection precision, and error propagation analysis

10. Critical-reading questions

Supervision boundary

The method does not use target-image labels for training, but it does use:

- known class names;
- LLM-generated class descriptions;
- pretrained CLIP and DINO models;
- an unlabeled target-image collection.

Call it **label-free target adaptation**, not “learning with no supervision of any kind.”

Transductive advantage

NoLA sees the target training collection during adaptation. A zero-shot CLIP baseline does not adapt to that collection. This is legitimate, but the distinction must be stated clearly.

Confidence bias

Top-k selection can create class-balanced training data, but it can also select confidently wrong images. Under severe domain shift, confidence may be poorly calibrated. The paper does not deeply measure this failure mode.

Hard pseudo-labels

The code takes the DINO-adapter softmax and then uses `argmax`, discarding uncertainty. Reliability weighting or soft targets may preserve useful information.

Experimental reporting

The paper reports one NVIDIA A100 but not runtime, memory, variance across seeds, or statistical significance. Reproduction should add these measurements.

Dataset-specific behaviour

NoLA slightly underperforms LaFTer on EuroSAT in Table 1 even though EuroSAT is one of our proposed core datasets. That makes EuroSAT scientifically useful: improvement is not guaranteed, and failure analysis can be meaningful.

11. Map the paper to the official code

Main file: `code/nola-official/trainers/NoLA.py`

Paper concept	Code location	What to inspect
Selected pseudo-label dataset	lines 35–65, `TopKDataset`	Stores pseudo-label, confidence, and ground truth for analysis
Frozen DINO and adapter	lines 82–112, `Taal` and `AdapterMLP`	DINO ViT-B/16 plus 256-unit hidden adapter
CDE generation	around lines 149–183, `gen_emb`	Loads or averages description embeddings
Visual prompts	around lines 232–247	Adds prompt embeddings to image patch tokens
Trainable parameters	lines 279–319, `setup_optim`	Prompts, normalisation layers, and CDE adapter
Pipeline startup	lines 419–456, `before_train`	Finds k, selects samples, trains and freezes DINO adapter
Prompt-training loss	lines 457–476, `forward_backward`	DINO hard pseudo-label from weak view; prompted CLIP on strong view
Confidence selection	lines 478–537, `select_top_k`	Best insertion point for our reliability method
Adapter training	lines 539–563, `train_taal`	Cross-entropy on selected DINO embeddings

Two implementation details deserve attention:

- `select_top_k` computes DINO embeddings **after** CLIP/CDE confidence selection. Our extension needs DINO features for the wider target set before final selection.
- `forward_backward` converts DINO probabilities to hard labels using `argmax`. Reliability-aware weighting can be added around this loss after selection experiments are stable.

12. How the paper leads to our research contribution

The paper already demonstrates that DINO features complement CLIP. However, DINO is used mainly **after** confidence-only sample selection. This leaves a natural question:

> Why not use DINO's visual geometry to decide which CLIP pseudo-labels are reliable in the first place?

Our proposed DINO-Neighbourhood Confidence uses two signals:

- semantic margin from CLIP/CDE;
- agreement of nearby samples in DINO feature space.

For image `i`:

`w_i = alpha` *`semantic_margin_i + (1 - alpha)`* `neighbour_agreement_i`

The first experiment must preserve the exact same number of selected images per predicted class. That isolates **selection quality** from **sample quantity**.

The clean causal comparison is:

- NoLA confidence-only top-k;
- DINO agreement only;
- combined DNC;
- random selection with the same class budget;
- DNC with shuffled DINO features.

Measure pseudo-label precision before measuring final accuracy. This reveals whether DNC actually selects cleaner supervision.

13. Recommended reading sequence

Pass 1 — 35 minutes: build the map

Read:

- abstract;
- Figure 2 and its caption;
- contribution bullets at the end of the introduction;
- Sections 3.2.1–3.2.3;
- Table 1;
- conclusion.

Write only three notes:

- problem;
- three-stage method;
- main result.

Do not inspect code yet.

Pass 2 — 60 minutes: understand mechanisms

Read Sections 3 and 4 carefully. For each stage, record:

- inputs;
- frozen components;
- trainable components;

- supervision signal;
- outputs;
- possible source of error.

Then redraw Figure 2 from memory.

Pass 3 — 50 minutes: challenge the evidence

Study Tables 1–7 and answer:

- where does NoLA fail to beat LaFTer?
- which ablation most directly supports using DINO?
- are all compared methods under identical supervision?
- what evidence about pseudo-label reliability is absent?
- what compute information would be needed for reproducibility?

Pass 4 — 45 minutes: connect paper and code

Open `NoLA.py` using the code map above. Follow one image through:

– CDE prediction;

– top-k selection;

– DINO embedding;

– adapter prediction;

– weak/strong augmentation;

– prompt-learning loss.

Do not try to understand every utility function. Track data shape and gradient flow.

14. Original-paper exercises with marking

Complete these after the fast theory passes. Use the original PDF while answering.

Exercise 1 — reconstruct Figure 2 (10 marks)

Without copying the figure, draw all three stages and label:

- CLIP text encoder;
- CDE classifier;
- confidence-based top-k selection;
- frozen DINO encoder;
- alignment adapter;
- weak and strong views;
- visual prompt tokens;
- final loss.

Marking: 1 mark for each correct component, 2 marks for correct information flow, and 1 mark for correctly distinguishing frozen and trained components.

Exercise 2 — Equation 1 dimensions (8 marks)

Assume 10 classes, 50 descriptions per class, and a CLIP embedding size of 512. State the shape after description encoding, class-wise averaging, and classifier concatenation. Explain why averaging descriptions might improve over a single class-name prompt.

Marking: 4 marks for shapes, 2 for the averaging explanation, and 2 for identifying that language variation can also introduce noise.

Exercise 3 — supervision audit (8 marks)

List every source of information available to NoLA and classify each as target labels, target images, text knowledge, or pretrained knowledge. Explain why "label-free" is accurate but narrower than "unsupervised."

Marking: 5 marks for a complete inventory and 3 for the distinction.

Exercise 4 — analyse Table 1 (10 marks)

Find:

– the 11-dataset average gain over CLIP;

– the average gain over LaFTer;

– two datasets where the method is especially strong;

– every dataset where LaFTer beats NoLA;

– one reason an average can conceal useful failure information.

Marking: 2 marks per item.

Exercise 5 — methodological critique (12 marks)

Write a 250-word review containing:

- one major strength;
- one minor strength;
- two limitations;
- one requested experiment;
- an accept/reject recommendation with confidence.

Marking: 2 marks per evidence-grounded point and 2 marks for a balanced recommendation.

Exercise 6 — design the DNC experiment (12 marks)

Design a fair EuroSAT comparison between confidence-only top-k and DNC. Specify the fixed sample budget, allowed use of labels, metrics, seeds, controls, and stopping rule.

Marking: 2 marks each for fairness, supervision discipline, primary metrics, reliability metrics, controls, and reproducibility.

Answer-check targets

Do not read these until attempting the exercises:

- Equation 1 encoded descriptions can be viewed as `[10, 50, 512]`; averaging yields `[10, 512]`.
- Table 1 reports 76.6 minus 64.7 = 11.9 percentage points from displayed rounded values, while the text reports 11.91; it reports a 3.6-point gain over LaFTer.
- LaFTer is above NoLA on EuroSAT and CIFAR-10 in Table 1.
- Labels must not influence selection, adapter fitting, prompt training, or hyperparameter choice under the declared label-free protocol.

15. What to put in your own paper notes

Use one page with these headings:

– **Problem:** one sentence.

– **Setting:** what information is and is not available.

– **Method:** one sentence per stage.

– **Loss:** teacher, student, and trainable parameters.

– **Evidence:** three numbers from Table 1 and one ablation.

– **Weakness:** confidence-only selection under domain shift.

– **Our change:** DINO-Neighbourhood Confidence.

– **Fair test:** fixed per-class budget plus pseudo-label precision and final accuracy.

This structure is suitable for supervisor discussions and later becomes the backbone of the related-work and method sections.

16. Compact presentation script

CLIP provides semantic class knowledge but can have weaker fine-grained visual features. DINO provides strong visual structure but lacks target class semantics. NoLA uses LLM-generated class descriptions and CLIP to pseudo-label confident target images, uses those examples to train a small classifier on frozen DINO features, and then uses that DINO-based classifier to supervise visual prompt tuning of CLIP. It improves the reported 11-dataset average from 64.7% for zero-shot CLIP and 73.0% for LaFTer to 76.6%. Its open weakness is that initial samples are selected by classifier confidence alone. Our extension tests whether agreement in DINO feature neighbourhoods can identify more reliable pseudo-labels and improve label-free adaptation under domain shift.

17. Final one-page checklist

Before saying you understand the paper, verify that you can answer all of these without notes:

- What does CLIP contribute?
- What does DINO contribute?
- Why are LLM descriptions used?
- What is trained in each of the three stages?
- Where do the pseudo-labels come from in Stage B and Stage C?
- Why are weak and strong image views needed?
- What exactly does “label-free” exclude and permit?
- What are the main Table 1 numbers?
- Which ablation proves DINO matters most directly?
- What result is missing about pseudo-label reliability?
- Where would DNC enter the code?
- How will labels be prevented from leaking into training?

If any answer is unclear, return to Figure 2, Equation 3, or the corresponding code-map row rather than rereading the entire paper.